

A Predictive Model for Hotel Booking Status and Cancellation Risk

Data-driven insights for better customer retention



Hotel Haven

Presented by:
Kuroyefa Leghemo

Date Presented:
May 16th, 2026



Hotel Haven

Outline

- Business Introduction
- Problem Statement
- Objectives
- Key Business Questions
- Data Description
- Methodology
- Insights
- Model Performance Evaluation
- Recommendations
- Conclusions



Business Introduction

- Hotel Haven is a luxury hotel chain with multiple locations that offers a wide range of services including rooms, meal plans, and parking services

Problem Statement



01

**Struggle with high
cancellation rate**

02

**Lost revenue and
inefficient resource
allocation**

03

**Limited insight into
cancellation reasons**

Objectives

- Clean and prepare the hotel booking dataset for analysis and modeling
- Explore booking data to understand patterns, and cancellation behavior
- Perform comprehensive analysis to identify key factors influencing booking cancellations
- Engineer meaningful features that improve analysis and model performance
- Develop a machine learning model to predict booking status accurately
- Evaluate model performance using appropriate metrics for classification
- Fine-tune the model to improve accuracy and reliability
- Generate actionable insights that support better customer retention and reduce cancellations

Key Business Questions

Questions Addressed

- How does cancellation compare to honored bookings in Hotel Haven?
- What is the financial impact of cancellations on Hotel Haven's revenue?
- Are higher-priced bookings cancelled more than lower-priced ones?
- Are expensive bookings made far in advance more likely to be canceled?
- Are guests booked into certain room types more likely to follow through with their reservations?
- Does the market segment through which a guest books influence their booking status at Hotel Haven?
- Is the number of special requests a reliable signal of a guest's intention to honor their booking?

Data Description

- Source of Data: 10Alytics
- Size : 36,285 rows x 17 columns (16 features and 1 target) bookings records
- Period : Feb 2015 - December 2018 (3 years, 11 months)
- Key Features : booking ID, number of adults, number of children, number of weekend nights, number of weeknights, type of meal, car parking space, room type, lead time, market segment type, repeat, P-C, P-NOT-C, average price, special requests, date of reservation, booking status

Methodology

CRISP DM approach adopted

Data Cleaning and Preprocessing

- Relevant libraries imported and the dataset(CSV file) was loaded into jupyter notebook
- Data exploration done to understand the data types, quality and structure
- Descriptive analysis performed to summarize key metrics
- 37 booking records were found to contain an invalid date; February 29, 2018 (not a leap year), a possible system bug or data entry error during that period. All affected records were corrected to February 28, 2018
- No missing or duplicated values
- Wrong data type corrected appropriately

Methodology

Exploratory Data Analysis

- Features engineering: revenue column generated to enhance analysis
- Comprehensive data analysis and visualizations were done to uncover patterns in factors influencing booking cancellations

Data Preprocessing (cont'd)

- Irrelevant and repetitive columns were dropped (P-C and P-not = repeated)
- Outliers were investigated and removed
- Encoding of categorical variables
- Scaling of numerical data with Robust Scaler (mild - heavy skewness)
- The target had class distribution imbalanced (67:33), which was handled using (Synthetic Minority Oversampling Technique (SMOTE))

Final Cross-Validation

Methodology

Modelling

- Trained Baseline Model – Logistic Regression model (78% prediction accuracy, F1-score and recall)
- Trained Multiple Advanced Classification Models
 - Random Forest (best performance)
 - Gradient Boosting
 - Ada Boost
 - SVC
 - K Nearest Neighbours
 - Decision Tree
 - XG Boost
- Hyperparametric tunings

Methodology

Tools

- Python (Jupyter Notebook)
- Libraries;
 - Matplotlib
 - Pandas
 - Seaborn
 - Scikit Learn
 - XG Boost
 - Imbalanced-learn
 - Dateutil

Hotel Haven Key Performance Metrics

36,285

\$7M

32.8%

\$4.2M

85 Days

**Total Hotel
Bookings**

**Total Revenue
Generated**

**Total
Cancellation
Rate**

**Total Revenue
Loss**

**Average Lead
Time**

Insights

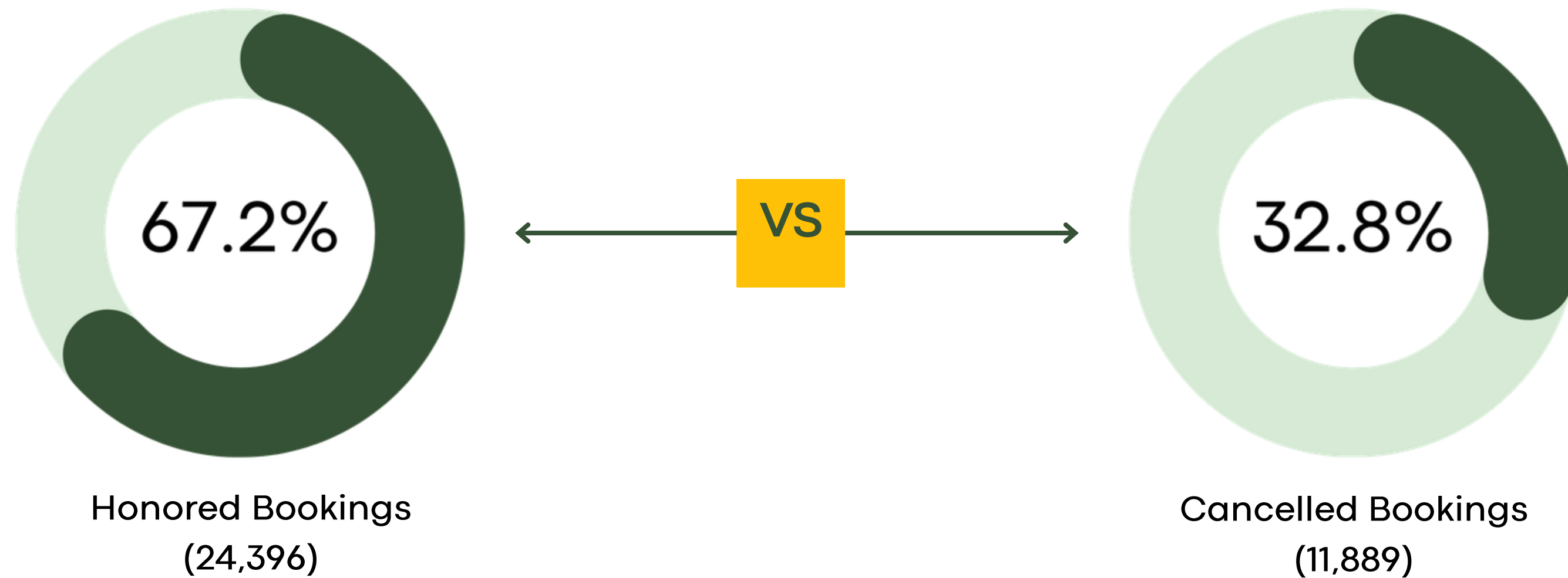


Figure 1: Booking Status Overview at Hotel Haven

Key Insights

The \$101-\$200 price range is Hotel Haven's biggest cancellation problem, with nearly 7,000 cancellations, representing the greatest source of lost revenue for the hotel

Hotel Haven attracts very few premium bookings (\$301-\$540)

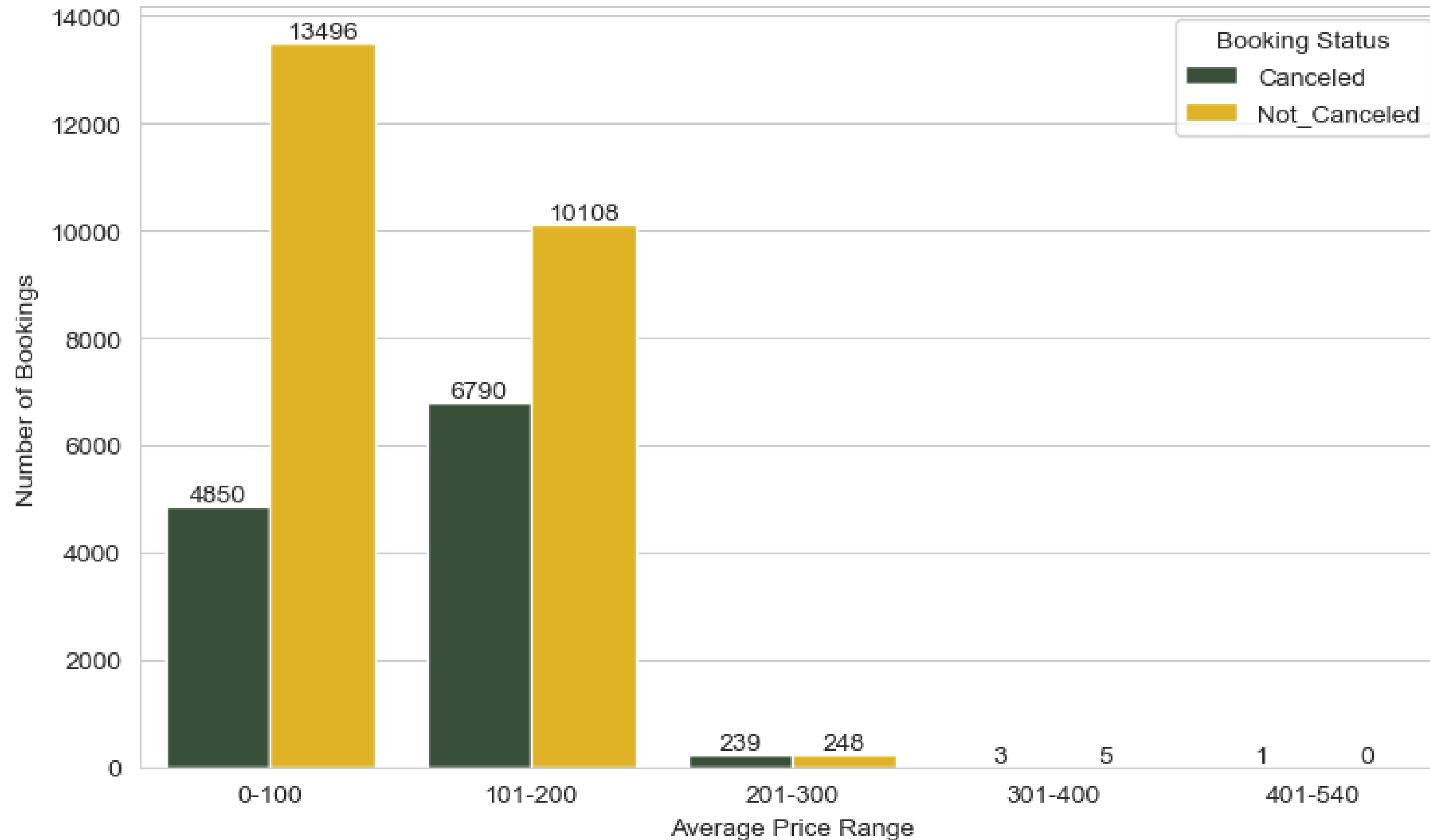


Figure 2: Distribution of Booking Status by Average Price

Key Insights

Lead time is the dominant cancellation driver, regardless of the price

Guests who book ≤ 30 days before arrival are far more likely to follow through

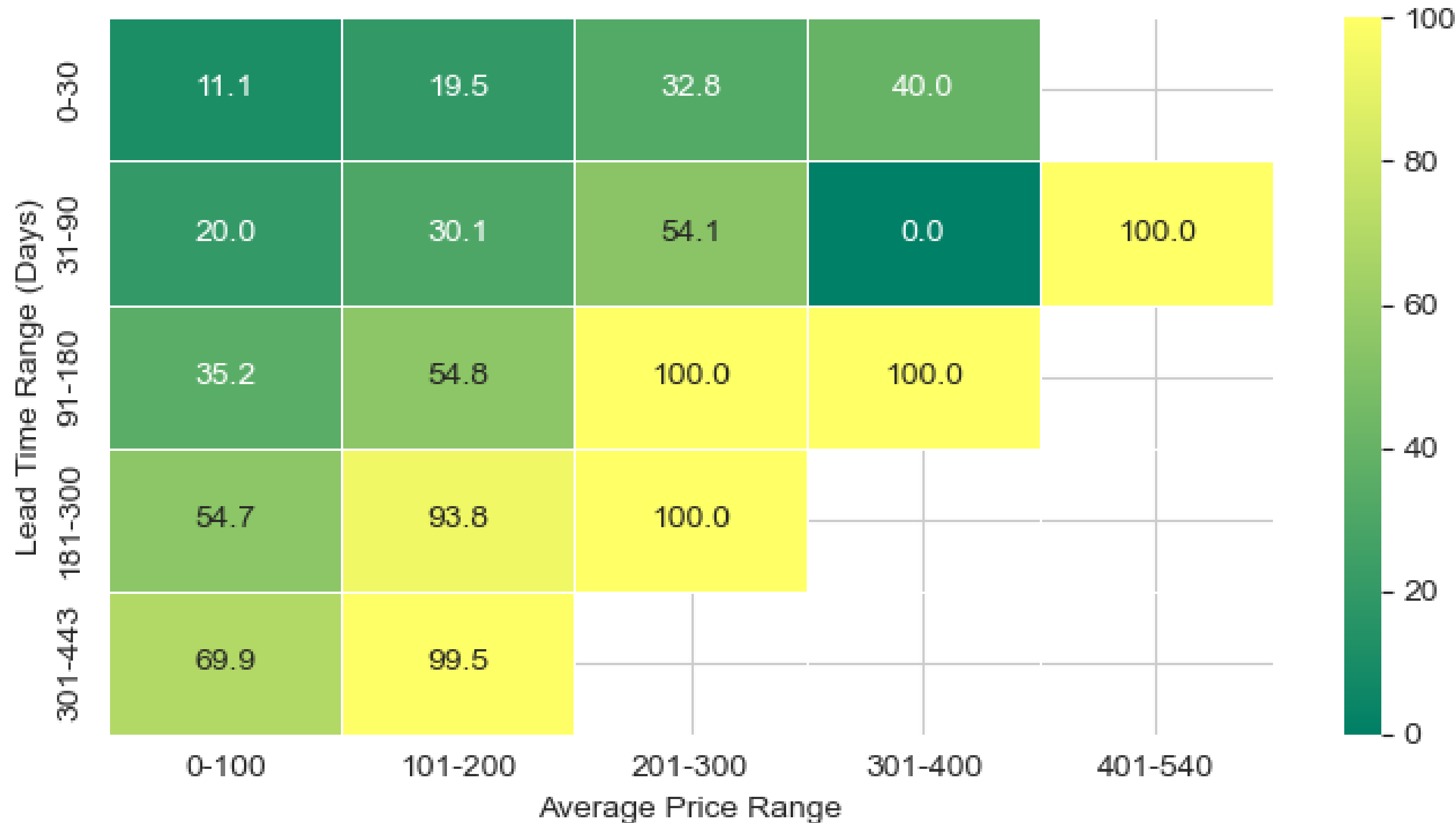


Figure 3: Cancellation Rate (%) by Average Price Range and Lead Time

Key Insights

No clear/consistent pattern of cancellation rate across room types

Slight variations suggests that room type alone is not a strong predictor of cancellations

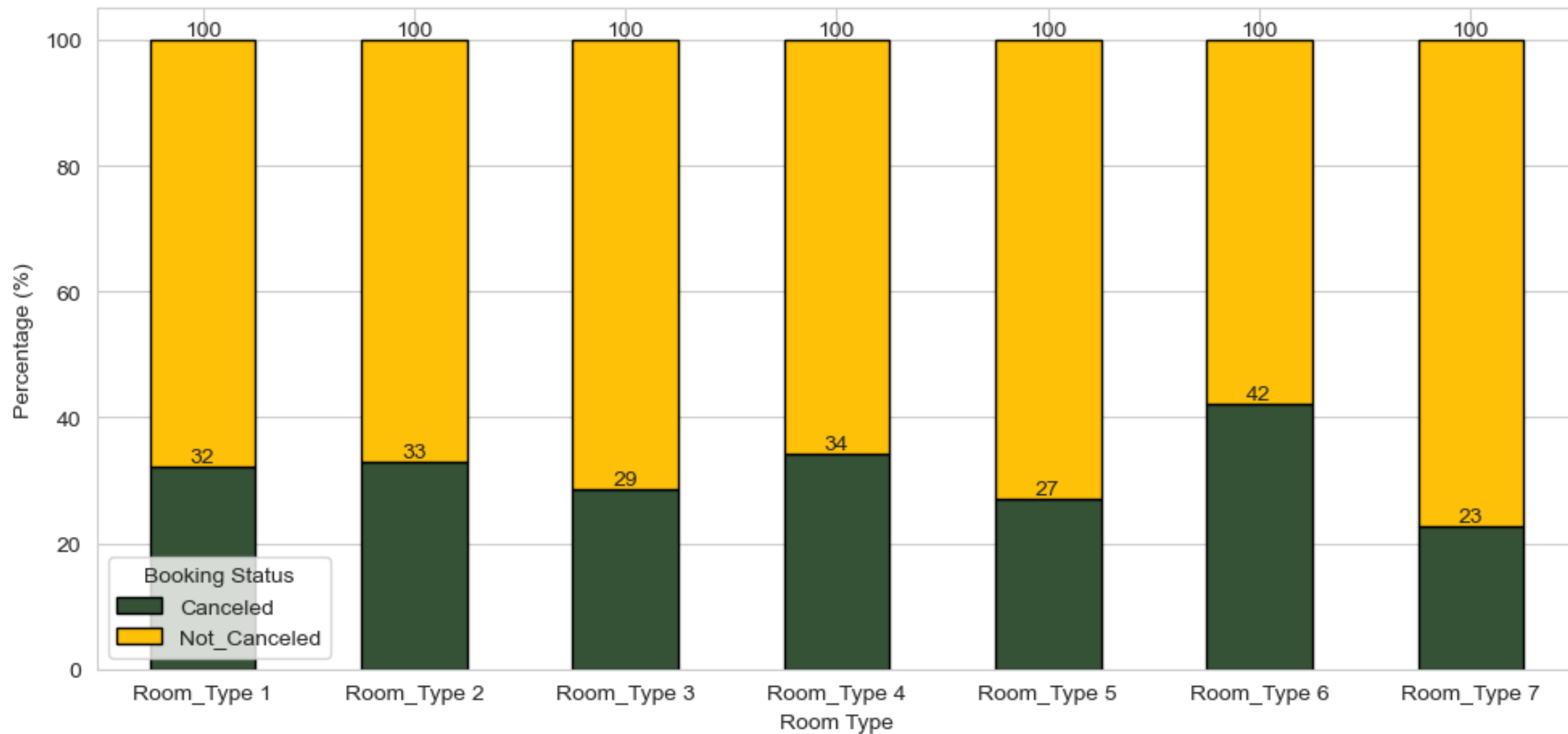


Figure 4: Percentage Distribution of Booking Status by Room Type

Key Insights

The seamless booking and cancellation of the online booking channel directly contributes to revenue and inefficient resource allocation at a cancellation rate of 37%.

With only 11% cancellation rate, corporate guests demonstrate strong booking commitment.

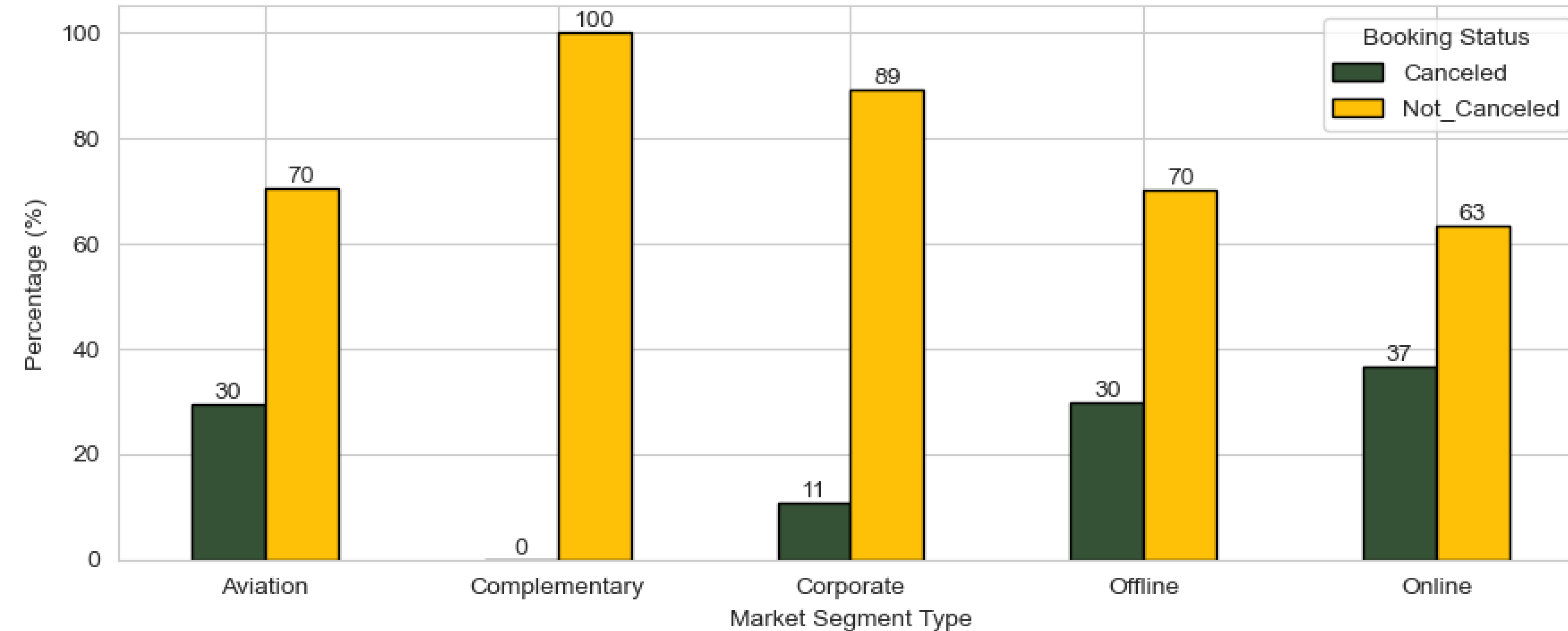


Figure 5: Percentage Distribution by Market Segment

Key Insights

Special request is a reliable signal of booking commitment. Each additional special request made at booking significantly reduces the likelihood of cancellation.

Guests with zero special requests cancel at 43%, making them Hotel Haven's highest cancellation risk segment

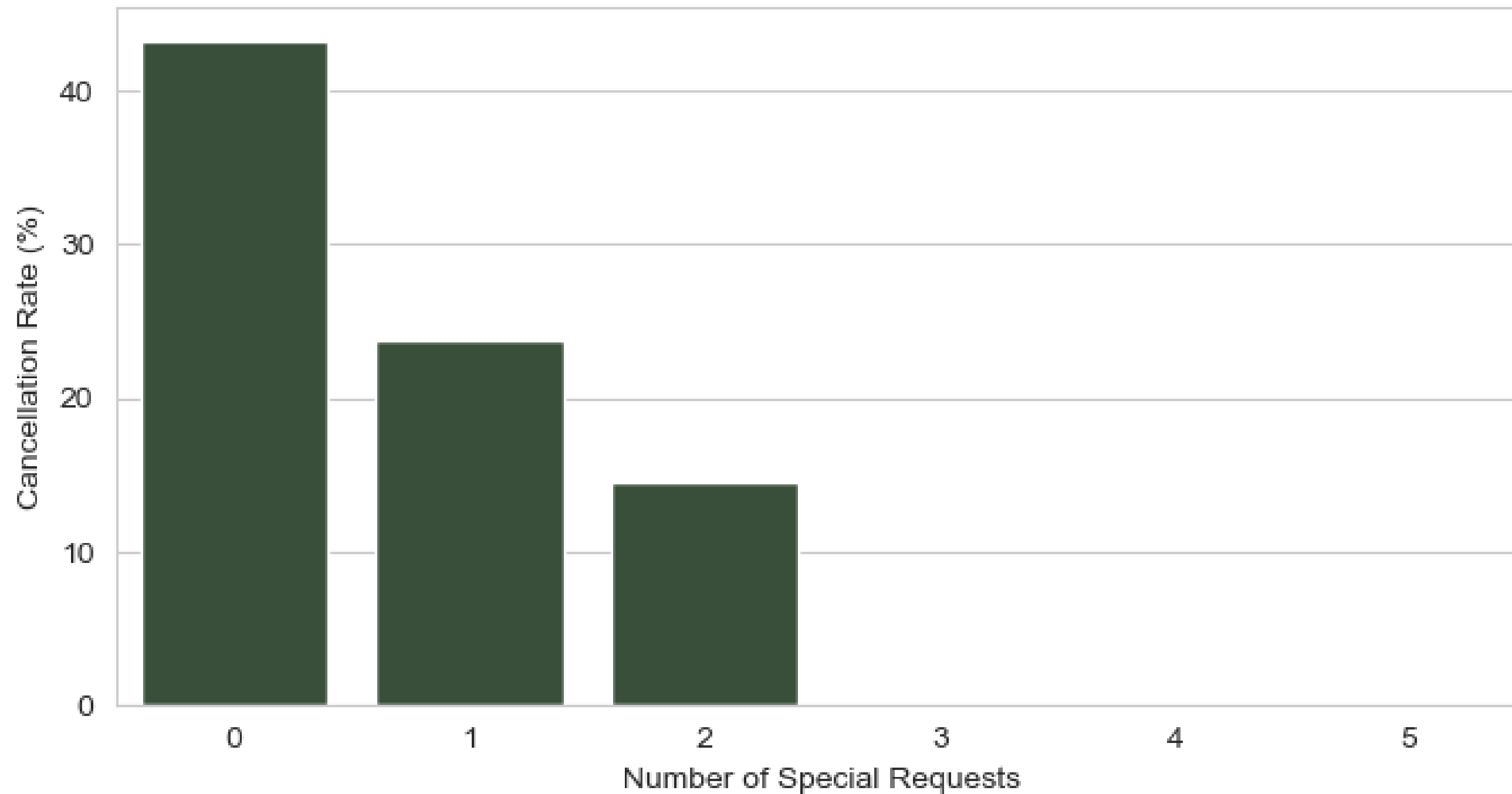
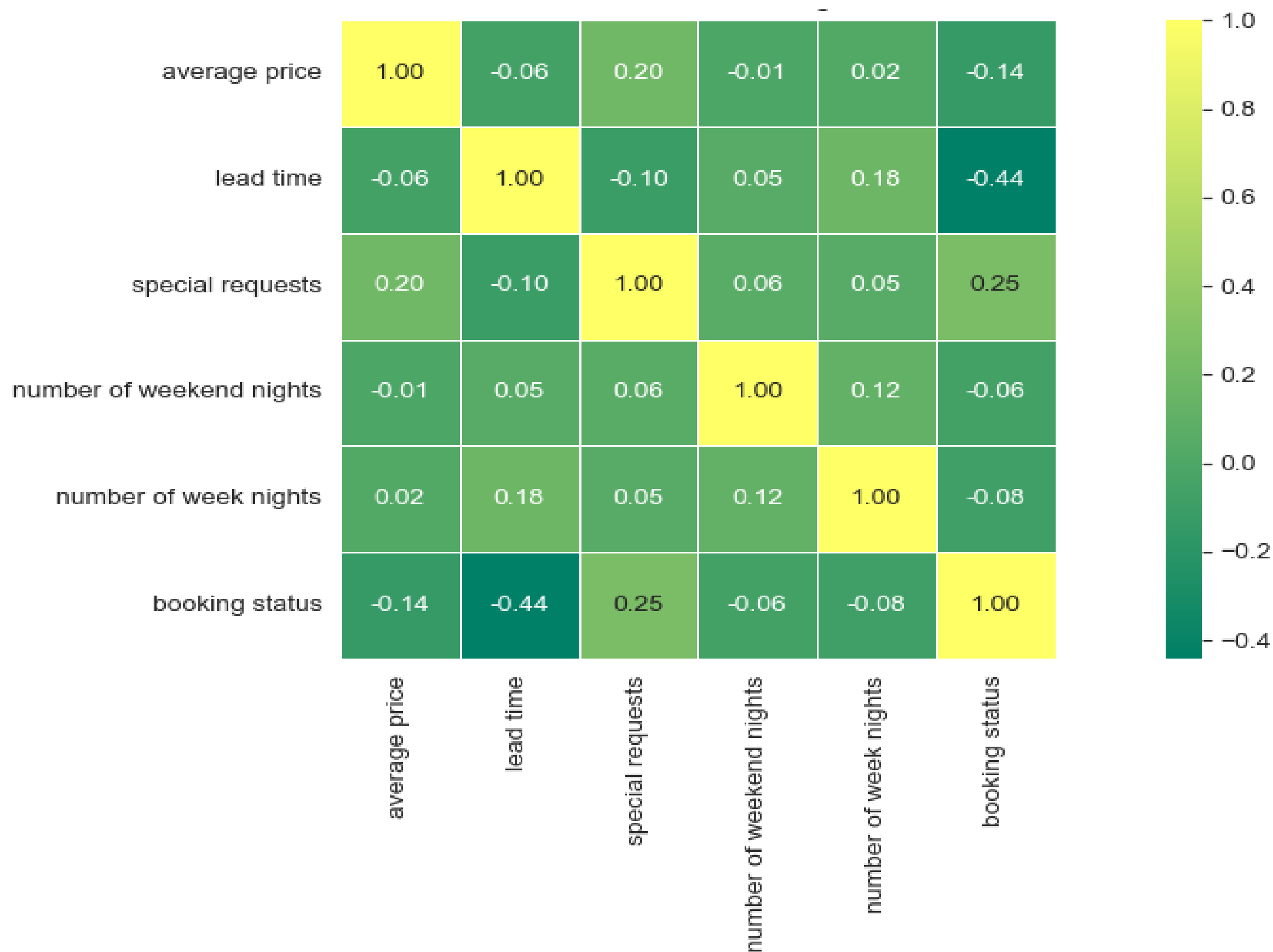


Figure 6: Cancellation Rate by Number of Special Requests



Key Insights

- Lead time (negative predictor), special requests and (positive predictor) are the factors which mostly influence booking status at Hotel Haven reinforcing our earlier finding.
- Price is a weak predictive factor

Figure 6: Correlation Matrix of Booking Variables

Model Performance Evaluation

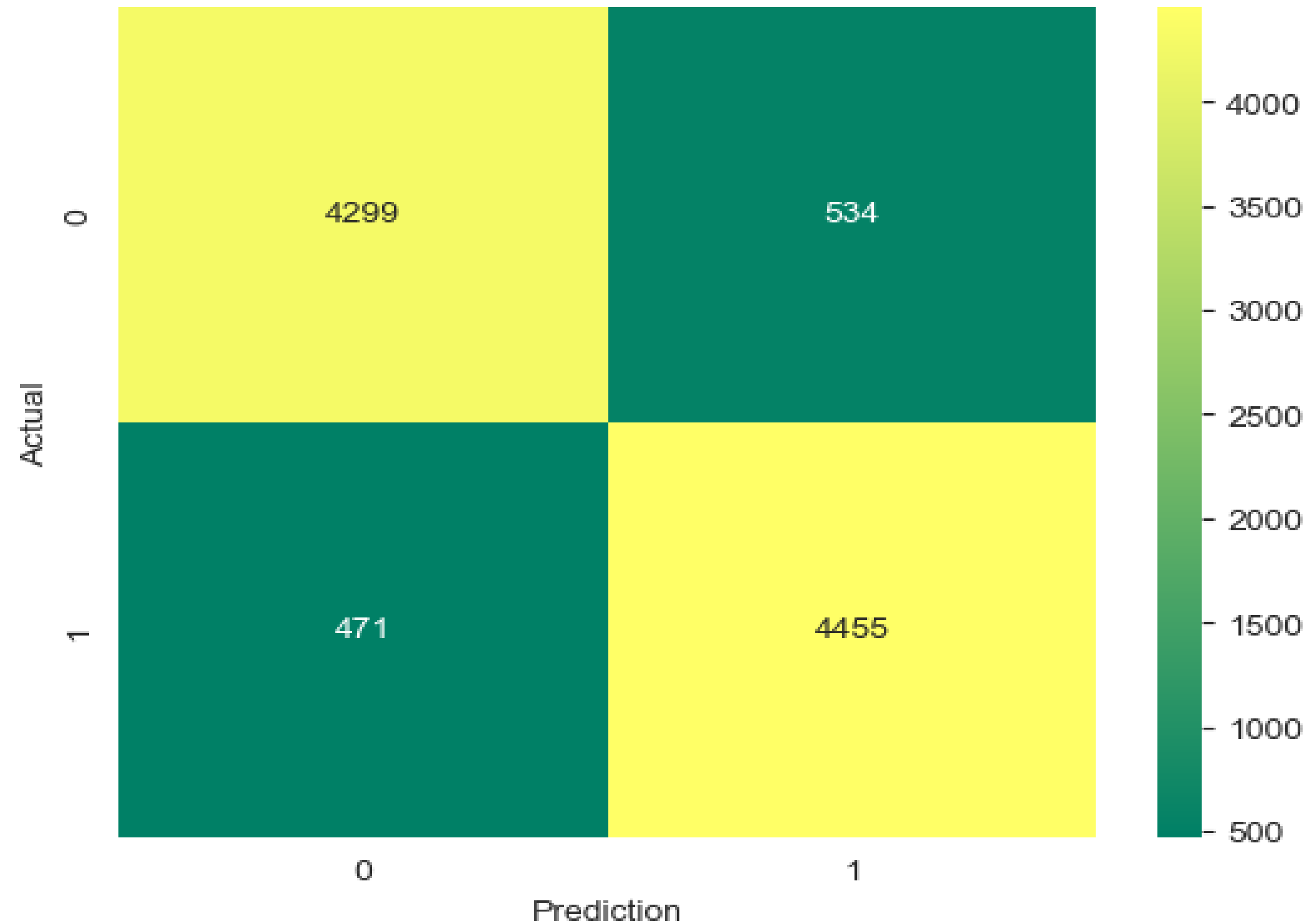


Figure 7: Confusion Matrix: For Random Forest Model

Model Performance Evaluation

Accuracy: 90% (correctly classified prediction)
F1 Score: 90% (balanced precision and recall)
Recall: 89-90% (correctly flags cancellation)
Precision: 89-90% (low false alarm rate)
ROC-AUC: 96% (strong class readability)

Recommendation

Price and Booking Commitments

- Use deposits, prepaid rates, or credit card to increase booking commitment
- Offer flexible options such as rescheduling or partial refunds instead of full cancellation for lower priced rooms, giving budget guests a financial incentive to commit at the point of booking

Market Segments

- Room upgrades or loyalty reward to guests who book directly (offline market segment)

Lead Time

- Identify long lead-time bookings at an early stage and proactively engage these guests with personalized campaigns, promotional upgrades and experience packages
- Offer small incentives for guests who keep long-lead bookings

Recommendation

Room Type

- Improve listing accuracy by ensuring room descriptions, photos, amenities, and prices match the actual experience

Market Segment

- Actively grow the corporate segment by proactive contracts negotiation and corporate rate agreements with companies, and travel agencies, as their accommodation partner for corporate travellers
- Incorporate more options in special-request fields

Conclusion

- Hotel cancellations is a major financial burden in the hospitality industry as it causes revenue loss, poor allocation of resources, and reduced profitability
- For Hotel Haven, this problem is especially significant, with a recorded revenue loss of \$4.2 million from a 32.8% cancellation rate over a 3 years, 11 months period
- While they are not the direct causes of cancellations, lead time, special requests and pricing are the strongest predictors of cancellations
- By leveraging machine learning models, Hotel Haven can make proactive, data-driven decisions that move the business from reacting to cancellations to preventing them, ultimately reducing cancellation rates, improving profitability, and strengthening overall business performance

Thank You

Reference

1. D-EDGE (2024). *Hotel Distribution Report*. Available at: <https://www.d-edge.com/avoid-guests-ghosting/>
2. HotelMinder (2024). *Reduce Hotel Cancellations: 8 Practical Tips*. Available at: <https://www.hotelminder.com/8-tips-to-reduce-last-minute-hotel-cancellations-and-no-shows>
3. Smart Order. (2025). *Hotel Minimize Hotel Booking cancellation*. Available at: <https://www.smartorder.ai/resources/blog/how-to-minimize-hotel-booking-cancellations/>